Visual Media Bias: Evidence from Computer Vision and Machine Learning

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In this project, I study how newspapers use visual content as a strategic tool of political persuasion. While most work on media bias focuses on text, recent evidence shows that images encode bias in both the selection of politicians and their portrayal, including through facial expressions and emotional cues. I examine a distinct channel: the selective use of photographs that are more or less visually flattering due to photography-dependent features.

These features include aspects such as lighting, camera angle, timing, posture, sharpness, and composition, which can amplify or conceal aesthetic imperfections without altering a politician's underlying physical traits. I hypothesize that newspapers systematically select less favorable images of politicians from the opposing party, using image choice itself as a subtle yet salient form of distortion. Unlike prior work that focuses on facial expressions or emotional content, which may be correlated with the tone or substance of the accompanying article, my approach isolates variation in visual representation that arises purely from how the photograph is taken and selected.

To identify this mechanism, I exploit within-politician variation during electoral campaigns. Specifically, I focus on the transition in which a candidate becomes the presumptive party nominee. This event sharply increases the politician's political salience – turning them into the primary target of opposing media – while leaving their underlying appearance essentially fixed. I combine this within-politician event with cross-newspaper ideological differences to estimate whether opposing outlets begin to use systematically less flattering photographs following nomination. This event-study design allows me to isolate strategic changes in image selection from fixed differences in politician characteristics or baseline media exposure.

Empirically, I construct a novel dataset linking newspaper articles to the images used to depict political candidates. I then develop a measure of visual favorability using computer vision techniques that capture photography-dependent features rather than emotional content. The resulting measure reflects how flattering or unflattering a politician appears due to the conditions under which the photograph was taken and selected.

By combining advances in computer vision with a causal empirical design, this project provides new evidence that media bias operates not only through what newspapers say, but also through how they choose to visually portray political actors. More broadly, it highlights a previously understudied channel through which media can shape perceptions in a subtle, low-cost, and difficult-to-detect way.